

Experiment 04 - Exercise 07: Fruit Sets (Intersection, Difference, Union)

Tofik Ali

February 17, 2026

Objective

Create two sets of fruits and find:

- fruits in both sets,
- fruits only in the first set,
- count of all unique fruits from both sets.

Input

Number of fruits n and fruit names for sets **s1** and **s2**.

Output

Intersection, difference, union, and the union size.

Approach

Use set operations:

- Intersection: **s1 & s2**
- Difference: **s1 - s2**
- Union: **s1 | s2**
- Count of all unique fruits: **len(s1 | s2)**

Time Complexity

$O(n)$ average-case for building sets and computing operations.